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Spacecraft Attitude Dynamics
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1 Project specifications

	Assigned specifications	Modifications	Motivation for modification
Platform	Microsat (10-100 kg)	-	-
Attitude parameters	Quaternions	-	-
Mandatory sensor	Sun Sensor	Added a magnetometer	A second measurement was needed to determine attitude
		Added a gyroscope	Allow to measure the spacecraft angular velocity after failure of implementing observer
Actuators	Variable thrust jets	-	-

Table 1: Project specifications

2 Mission definition

The primary objective of this mission is to control the attitude of a micro-satellite in a sun-synchronous orbit at an altitude of 1,622 kilometers. The spacecraft will keep consistent sun-pointing for optimal solar panel performance. The mission requirements have been designed to address these objectives while considering the relevant environmental disturbances and operational constraints.

2.1 Spacecraft Configuration

The spacecraft is in the micro-satellite category, with a mass of 50 kilograms. Its compact size makes it cost-effective while ensuring it is capable of fulfilling the mission objectives. The attitude control system leverages quaternions to represent and compute the spacecraft's orientation, ensuring robust and singularity-free calculations.

2.1.1 Size

The spacecraft's mass and dimensions were selected to align with the mission objectives. The mass of 50 kg places the spacecraft in the mid-range of micro satellites (10 to 100 kg), providing a balance between sufficient payload capacity for advanced sensors and actuators, and low launch costs.

The dimensions have been chosen to make it compact (0.3m×0.2m×0.1m) and not symmetric, mainly for inertia moment reasons, explain below. These dimensions also ensure efficient thermal management during sun-pointing phases, reducing exposure and thermal fluctuations on critical components. A compact satellite also reduces the potential residual drag force, which will be very limited at the orbit altitude, but affect the mission on a long term.

The moments of inertia were calculated based on the spacecraft's mass and dimensions, which results in $I_x = 0.183; I_y = 0.267; I_z = 0.342\ kg.m^2$. These values ensure rotational stability and manageable torque requirements. The moderate values of inertia allow for precise attitude adjustments without requiring excessive energy from the attitude control system.

2.1.2 Sensors

Several sensors are used on board to perform the attitude control :

1. **Sun Sensor**: Used to determine the direction of the Sun relative to the spacecraft. While this sensor provides critical information for sun-pointing, it alone is insufficient for fully determining the spacecraft's attitude.
2. **Magnetometer**: Measures the Earth's magnetic field, providing additional data to complement the Sun sensor. This allows to resolve ambiguities in the spacecraft's orientation.
3. **Gyroscope**: Measures angular velocities, offering real-time data to support precise attitude determination and dynamic maneuvers. The gyroscope was added because of observer implementation failure.

2.1.3 Actuators

The attitude control is achieved using variable thrust jets, which provide fine control of the spacecraft's orientation by generating magnitude controlled torques. Jet thrusters have different configurations in regards to their orientation with respect to each other and their placement with respect to the center of mass of the spacecraft, which will determine the net torque produced by the thruster. As a preliminary design, a simple 4 thrusters configuration was selected, like the one shown in the image: by searching for available commercial jet thruster, the specifications for

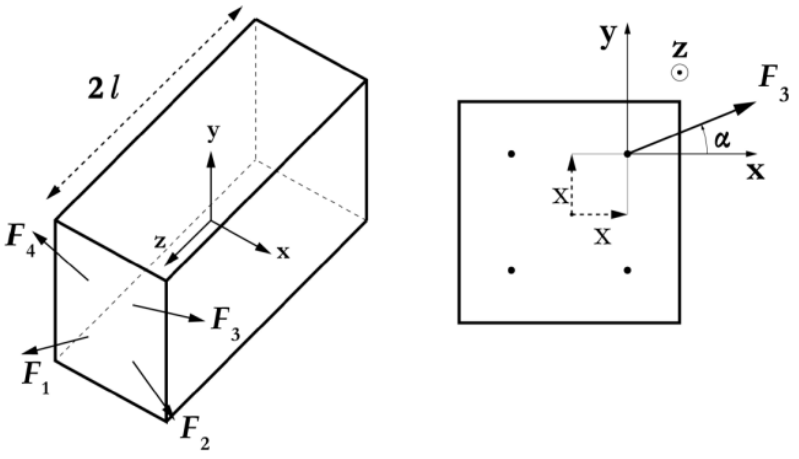


Figure 1: Selected thruster configuration, figure obtained from (1)

the thrusters were set as follows:

- Minimum impulse bit: 0.5 mN-sec
- Thruster valve response time: <20 ms
- Maximum single thruster response: 25 mN

2.2 Attitude Control Objectives

The spacecraft is designed for 3-axis stabilization purposes, ensuring precise control over its orientation to meet mission objectives, i.e. Sun pointing. The spacecraft aligns its solar panels towards the Sun to maximize power generation. This mode ensures continuous operation of the satellite's systems.

2.2.1 Operational modes

The attitude control system supports the following operational modes:

1. **De-Tumble**: The spacecraft stabilizes itself from an initial uncontrolled state by reducing angular velocities to near-zero values. Note that the initial state is guaranteed by the launcher to be relatively small after the separation process.
2. **Slew Maneuver**: This maneuver re-orients the spacecraft to the specified orientation, in our case it orients the spacecraft to point towards the sun. It ensures smooth and controlled reorientation to achieve the desired attitude.
3. **Pointing**: The spacecraft maintains its 3-axis stabilized in a particular orientation.

2.2.2 Environmental Disturbances

To ensure accurate attitude control, the following disturbances are taken into account for designing the system:

1. **Gravity Gradient Torque**: Caused by the Earth's gravity field acting differently on different parts of the satellite. This torque depends on the spacecraft's orientation relative to Earth.
2. **Solar Radiation Pressure**: Results from the momentum transfer of photons from the Sun impacting the spacecraft's surfaces. This pressure introduces small but continuous forces and torques.
3. **Magnetic Torque**: Induced by the interaction between the Earth's magnetic field and the spacecraft's magnetic moment. This disturbance varies with the satellite's position in orbit and must be counteracted for precise attitude control.

2.3 Mission Orbit

2.3.1 Orbit Selection

A sun-synchronous orbit was chosen as it removes eclipse periods, ensuring consistent exposure to sunlight. This is a major criterion for fulfilling the sun-pointing requisite.

2.3.2 Orbit Parameters

- **Altitude**: 1,622 kilometers. This altitude was selected to mitigate atmospheric drag, which is considered as being negligible at this height.
- **Inclination**: 98 degrees, characteristic of Sun-synchronous orbits, that the orbit never goes behind Earth.

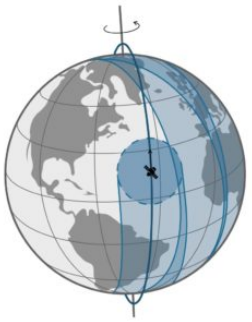


Figure 2: Illustration of a Sun-Synchronous Orbit

2.4 Summary

The mission aims to achieve reliable and precise attitude control of a micro-satellite in a sun-synchronous orbit, utilizing a set of sensors and actuators. The system is designed to counteract significant environmental disturbances while fulfilling the objective of sun-pointing. The proposed orbit and attitude control strategies are designed to maximize the mission's effectiveness and operational longevity.

3 Simulation model

3.1 Sensors

The model of the sensors has been done in a way that an ideal sensor is developed first and, once its smooth operation has been verified, some relevant error sources are added to it.

For both sun and magnetic sensors, the errors come from :

- **Bias error :** Refers to a constant offset present in the sensor's output, regardless of the input value. This can occur due to imperfections in the sensor's electronics or manufacturing process.
- **Internal misalignment :** Occurs because the sensor's axes are not perfectly aligned with the reference coordinate system of the spacecraft. This can happen due to assembly tolerances or mechanical deformations.
- **Scale factor error :** Represents inaccuracies in the sensor's gain for amplifying the input signal.
- **Digitalization process :** Arise during the conversion of the analog sensor output to a digital signal. This process involves quantization, where continuous input values are rounded to discrete digital levels. The finite resolution of the analog-to-digital converter introduces small errors, particularly for low-amplitude signals.
- **Time Delay :** Refers to the lag between the actual occurrence of a measured event and the time the measurement is available for use in the system. This delay results from the sensor's internal processing.

Finally, a random error source is added. For the sun sensor and magnetometer noise, a colored noise is used instead of a white noise to better represent real-world conditions. Unlike white noise, which assumes random and uncorrelated errors, colored noise captures the frequency-dependent characteristics of sensor errors. For the sun sensor, colored noise accounts for slow-varying

disturbances like light variations and aging effects. For the magnetometer, it models frequency-dependent magnetic field fluctuations and electromagnetic interferences. This approach enhances the realism of the simulations and ensures more accurate performance predictions for attitude determination and control systems.

As a gyroscope is affected inherently to the high frequency disturbances, it is more relevant this time to model its random noise error by a white noise. This implies to be in an unbiased model, because a white noise leads to a zero-mean property (the random errors in measurements average out over time). Therefore, the error sources for the gyroscope are the Digitalization process and the time delay, which is a quite critical parameter here because the measurements are used to compute the control actions to apply on the spacecraft.

The above-mentioned errors have been added by modeling the sensors after real ones (2) (3) (4).

3.2 Dynamics and Kinematics

Two sub-programs are in charge of computing the spacecraft dynamic sand kinematics.

The angular velocity at time t is determined from the previous angular velocity at time $t-1$ and the total torque U currently applied to the spacecraft (command + disturbances) through the Euler equations.

When the angular velocity at time t is estimated, the quaternions are then determined from it and the quaternions value at time $t-1$ through the relation $\dot{q} = \frac{1}{2}\Omega(\omega)q$.

3.3 Environment

A sub-component Environment is here to compute/simulate the spacecraft velocity and position along the orbit. A magnetic field model is implemented to determine the Earth's magnetic field at the spacecraft's location. This is essential for simulating the data obtained by the magnetometer. Finally, the sub-component calculates the Sun direction vector relative to the spacecraft at each point in the orbit. This vector represents how the spacecraft is illuminated in real-world conditions, which allows to simulate the data obtained by the sun sensor.

All of these data allow to generate the three disturbances considered : gravity gradient, solar radiation pressure and magnetic torque.

3.4 Attitude Determination

The Attitude Determination block calculates the spacecraft's orientation, expressed as quaternions, by integrating data from the Sun sensor, magnetometer, and the spacecraft's position and velocity along its orbit. This block employs a statistical method that accounts for the varying accuracy of sensor measurements by weighting them based on their respective committed error.

The process begins with the Sun sensor providing the Sun direction vector relative to the spacecraft and the magnetometer measuring the local magnetic field vector. These measurements are combined with the spacecraft's position, obtained from the Environment block, to correlate the sensor data with known models of the Sun's position and Earth's magnetic field.

Using a weighted error statistical algorithm, the block assigns higher confidence to measurements with lower associated errors (sun-sensor). The algorithm then determines the spacecraft's

orientation (in terms of quaternions) by minimizing a cost function, called weighted error function, which is composed of the weighted measurements and the inertial reference vectors.

The choice of using a weighted error statistical algorithm is primarily motivated by its robustness and accuracy in handling sensor measurements with varying error characteristics. Unlike algebraic methods, which often assume ideal conditions and may neglect sensor inaccuracies, statistical methods explicitly account for sensor errors by assigning weights based on their noise characteristics. This ensures that measurements with lower reliability have a reduced influence on the computed attitude, leading to more accurate and stable results.

Compared to geometrical methods, which rely on deterministic relationships between sensor vectors and spacecraft orientation, statistical approaches are more adaptable to noisy and uncertain environments. Geometrical methods may fail or yield suboptimal results if the sensor data is significantly perturbed by errors. This makes the weighted error statistical algorithm a superior choice for real-world applications where sensor imperfections are unavoidable.

3.5 Controller and Thrusters

The Controller block is responsible for computing the ideal torque to be applied by the spacecraft's thrusters to achieve the desired orientation. Using the attitude information provided by the Attitude Determination block, the controller calculates the error between the current and desired orientations. The control logic is different for each modes. This approach (compared to using a single control logic) allows to have a better control depending on the use case.

The ideal torque computed is then sent to the spacecraft's thrusters, which generate the doable forces to adjust the orientation. This control loop ensures precise and reliable attitude adjustments, enabling the spacecraft to perform its stabilization objectives efficiently.

3.5.1 De-tumbling

The computed error is then processed through a PID (Proportional-Integral-Derivative) controller to determine the corrective torque.

The PID controller operates by combining three control actions:

- **Proportional (P) Control:** This component generates a torque directly proportional to the orientation error, ensuring a fast response to deviations.
- **Integral (I) Control:** By integrating the orientation error over time, the integral action corrects for steady-state errors, ensuring the spacecraft aligns perfectly with the desired attitude (i.e. steady-state error = 0).
- **Derivative (D) Control:** This component anticipates future errors by considering the rate of change of the orientation error, thus improving system stability and preventing overshoot. The Derivative action is also responsible for limiting or canceling oscillations in the response.

The advantages of using a PID controller include its simplicity, robustness, and versatility. It is well-suited for attitude control as it effectively balances the speed, accuracy, and stability of the response.

3.5.2 Slew Maneuver and Inertia Pointing

In these use cases, the control is computed from the Lyapunov's second stability theorem. This approach ensures at least the asymptotically stability of the system during the whole correction

process. To do that, a Lyapunov's control function is computed from the quaternions and the angular velocities. From this function, a control command is deduced, which ensures the spacecraft's correction while keeping the system stable.

3.5.2.1 Slew Maneuver

After detumbling the spacecraft to a desired angular velocities (could be just zeros), the spacecraft needs o re-oriented towards a target pointing before it can start its tracking mission. The control law for quaternions-based slew maneuver is obtained from (5):

$$u = -k_p \text{sign}(\delta q_0)\delta q_{1:3} - k_d \omega \tag{1}$$

where u is the desired toque, and $\delta q = [q_0 \quad q_{1:3}]$ is the attitude error obtained from $\delta q = q \otimes q_{desired}^{-1}$, this control law provides the shortest path to the desired attitude, due to the term $\text{sign}(\delta q_0)$ which will change the sign of the feedback in case $\delta q_0 < 0$.

3.5.2.2 Inertia Pointing

Now that the spacecraft is oriented towards our desired attitude, we may want it to track a moving target or point to an inertial position throughout its period motion while resisting the disturbances torques. The control law driving the motion for this part is also obtained from (5):

$$\mathbf{u} = [(\delta A \boldsymbol{\omega}_d)\times] I \delta A \boldsymbol{\omega}_d + I \delta A \dot{\boldsymbol{\omega}}_d - k_p \delta \mathbf{q}_{1:3} - k_d \delta \boldsymbol{\omega} \tag{2}$$

where δA is the attitude error obtained as $\delta A = A(q)A^T(q_d)$, and $\delta \omega = \omega - \delta A \cdot \omega_d$

3.6 Summary Scheme

The simulation model can be summarized into a scheme, shown below.

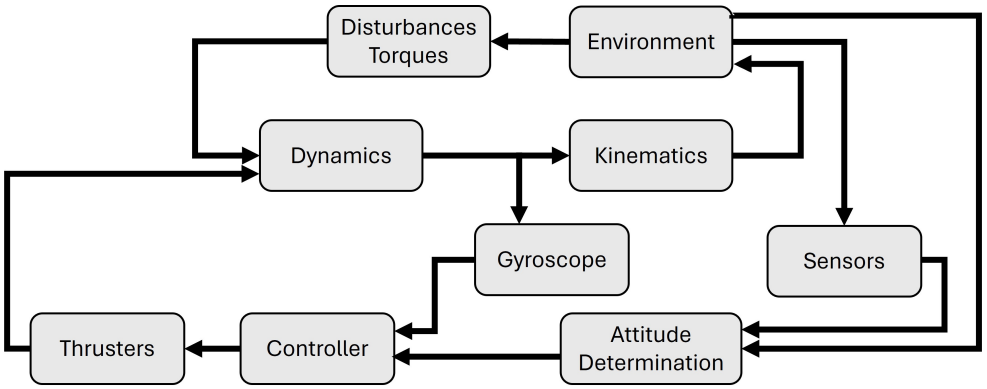


Figure 3: Simulation scheme of the Attitude Control Process

4 Results

The simulation model described has been used to compute the three main use cases. Note that as the initial configuration of the spacecraft is randomly selected each time the simulation is run, the reader may not found the exact same results as shown in this document.

4.1 De-tumbling

The simulation results focus on the de-tumbling mode, which represents the simplest use case for spacecraft attitude control. This scenario was specifically chosen as a first step to validate the simulation model. In this mode, the spacecraft starts with an arbitrary angular velocity and the control system is tasked with bringing it to a stable, non-rotating state. The results are given in fig. 4.

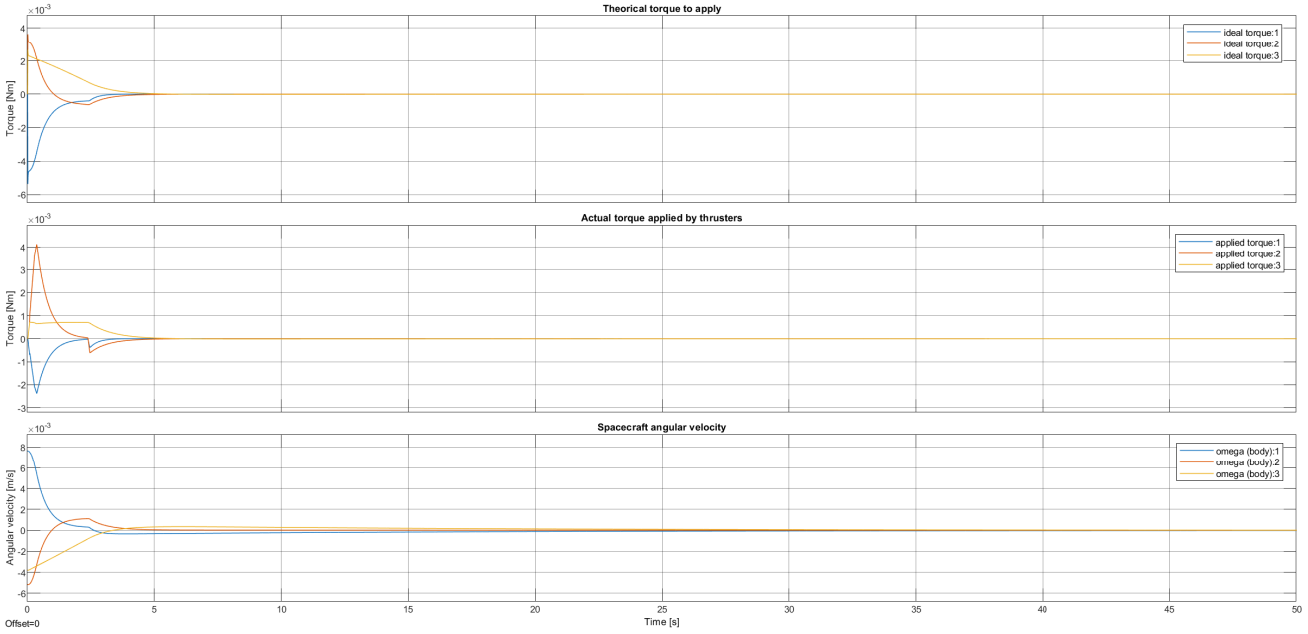


Figure 4: Ideal and real torques applied by the thrusters on the spacecraft *(zoomed on the first 50 sec for visibility)*

The results align well with expectations. The spacecraft successfully stabilizes within approximately 25 seconds, which demonstrates the effectiveness of the attitude control system in counteracting initial angular velocities, and thus despite external torques acting on the spacecraft. One notable observation is the slight difference between the ideal torque to apply computed by the controller, and the actual torque applied by the thrusters. This error is normal and arises from the practical limitations of actuator precision and the control loop dynamics. The error value is always small during the simulation (fig. 5) and quickly converges to 0.

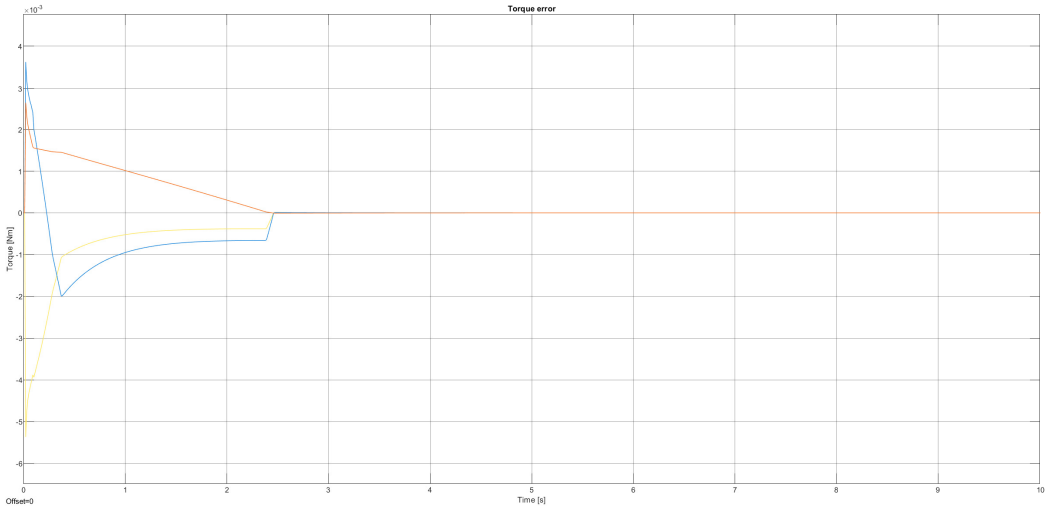


Figure 5: Torques error between the theoretical and actually applied (zoomed on the first 10 sec for visibility)

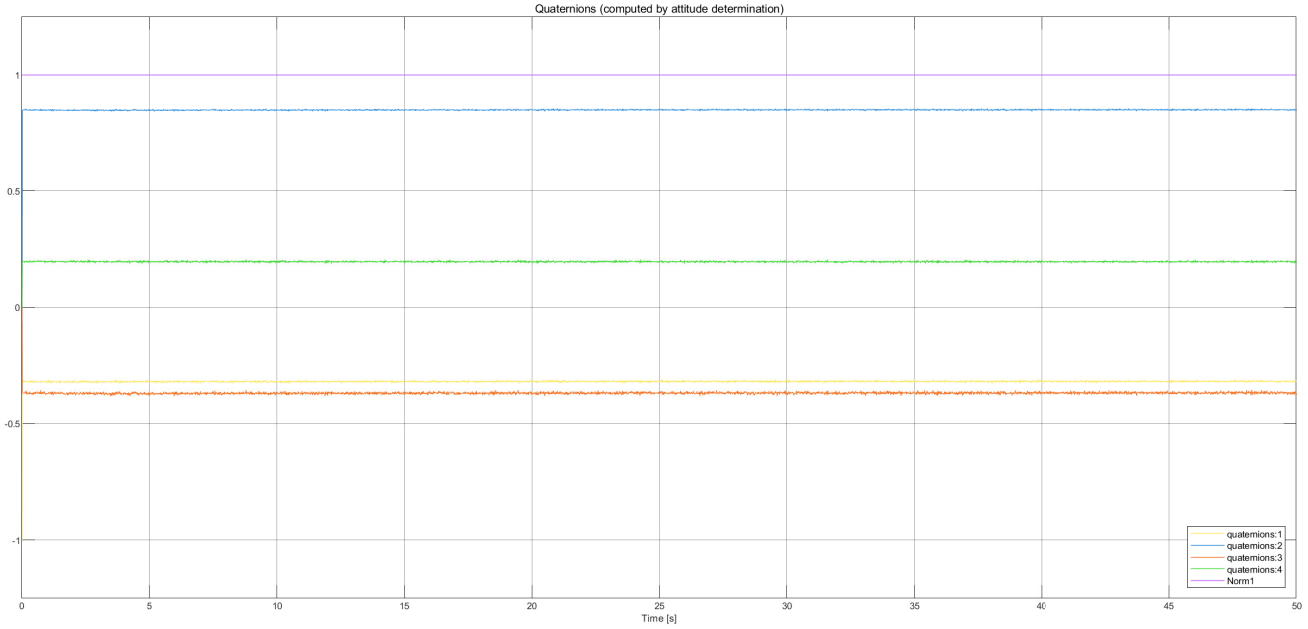


Figure 6: Quaternions determined by attitude determination block *(zoomed on the first 50 sec for visibility)*

Additionally, the computed quaternions (fig. 6) confirm the good computation of the elements during the simulation. Throughout it, the norm of the quaternion vector remains constant and equal to 1, as expected by the theory. This is a key indicator that the attitude representation is both stable and valid over time, further supporting the validity of the simulation model.

The quaternions obtained from the attitude determination process show slight oscillations. They are due to the inclusion of external disturbance torques, and show that they are effectively set into account. By accounting for these external influences, even if they are small (see fig. 7), the attitude determination algorithm provides a more accurate representation of the spacecraft's orientation in real-world conditions. The oscillatory behavior reflects the real-time adjustments made by the control system to counteract these torques and maintain stability.

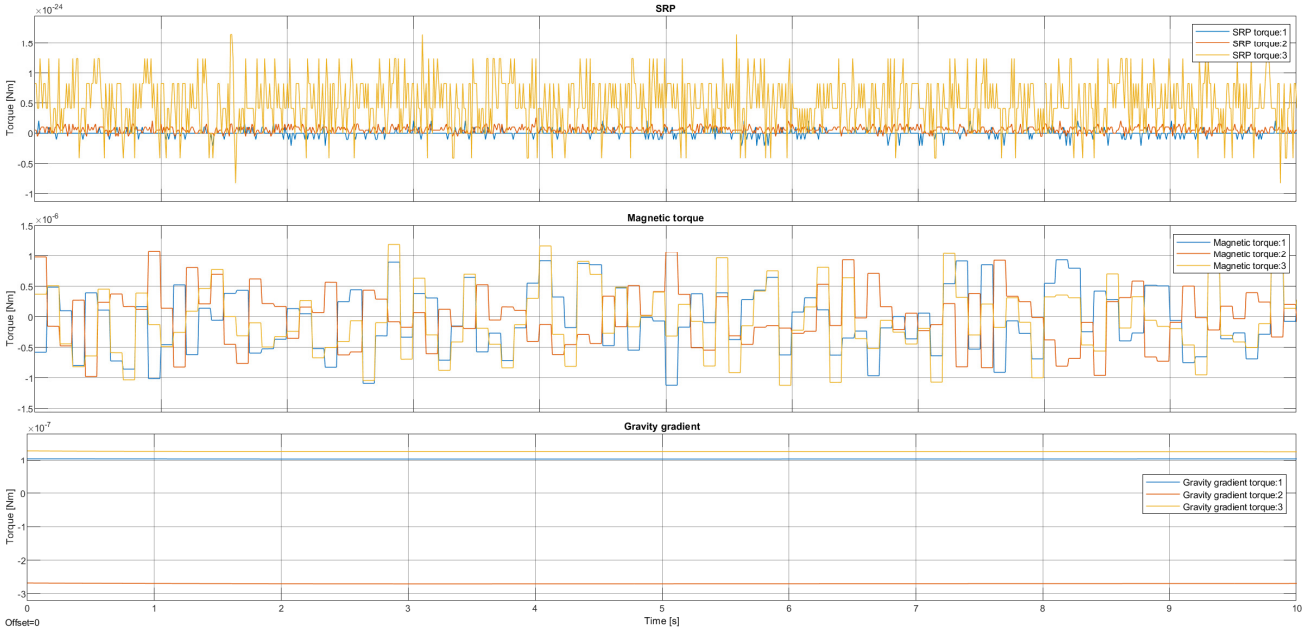


Figure 7: Disturbances torques acting on the spacecraft *(zoomed on the first 10 sec for visibility)*

Therefore, we can conclude by saying that the de-tumbling mode works as expected and validates the simulation model. Its allow us to confirm that the implemented algorithms, sensors, and actuators behave as intended under realistic conditions. The model is therefore ready to simulate the other and more complex modes.

4.2 Slew maneuver and Inertia Pointing

For this part and the following, the desired attitude corresponds to an angular velocity of $\omega_c = [0, \quad 0, \quad -n]$, where n is the orbital angular velocity. The target attitude is not related to the orbit, rather, it is just a demonstration of the controllers ability to track a specific attitude. The required quaternions that corresponds to this angular velocity is obtained from integrating the following formula:

$$\dot{q} = \frac{1}{2} q \odot \omega = \frac{1}{2} \mathcal{E}(q)\omega \tag{3}$$

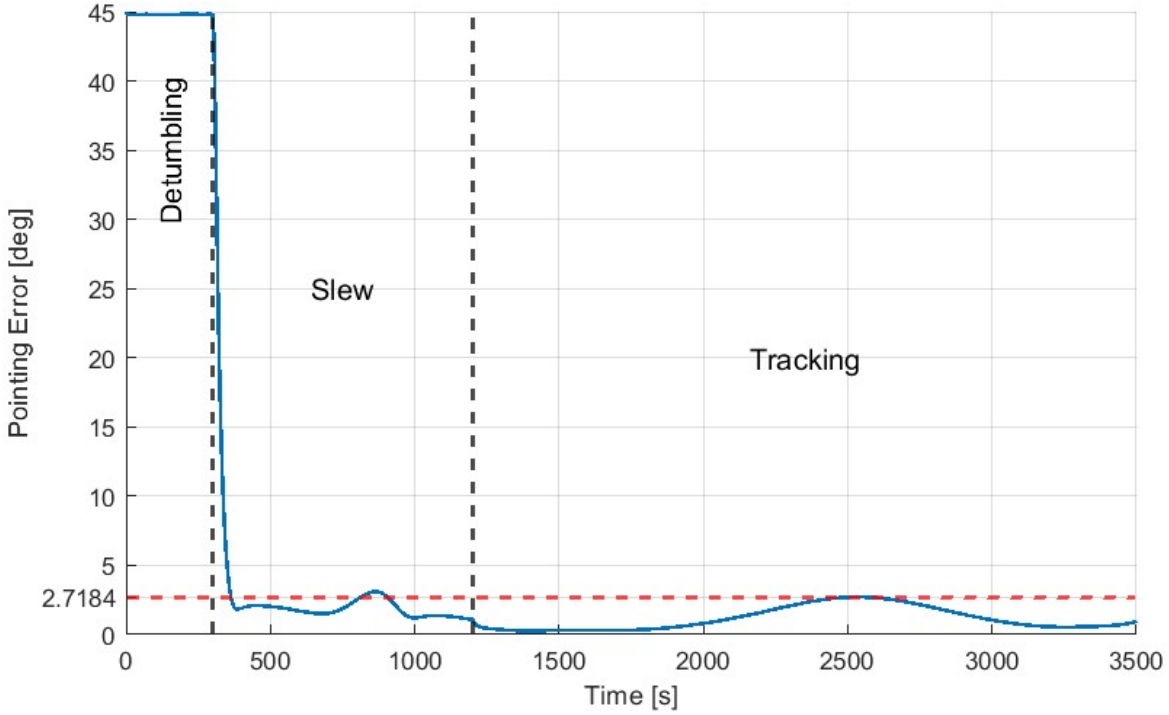


Figure 8: Pointing error throughout the different phases of orbiting.

As shown, the maximum pointing error during the tracking phase is 2.72 degrees. in the detumbling phase, there were no pointing requirements, only constrains on the angular velocity were imposed. Then, in the slightest part, the spacecraft started to orient itself with the required orientation. This value is might be a little large for application requiring high pointing accuracy. This error is obtained with the presence of 3 different environmental disturbance torques, and uncertainties in sensors measurements.

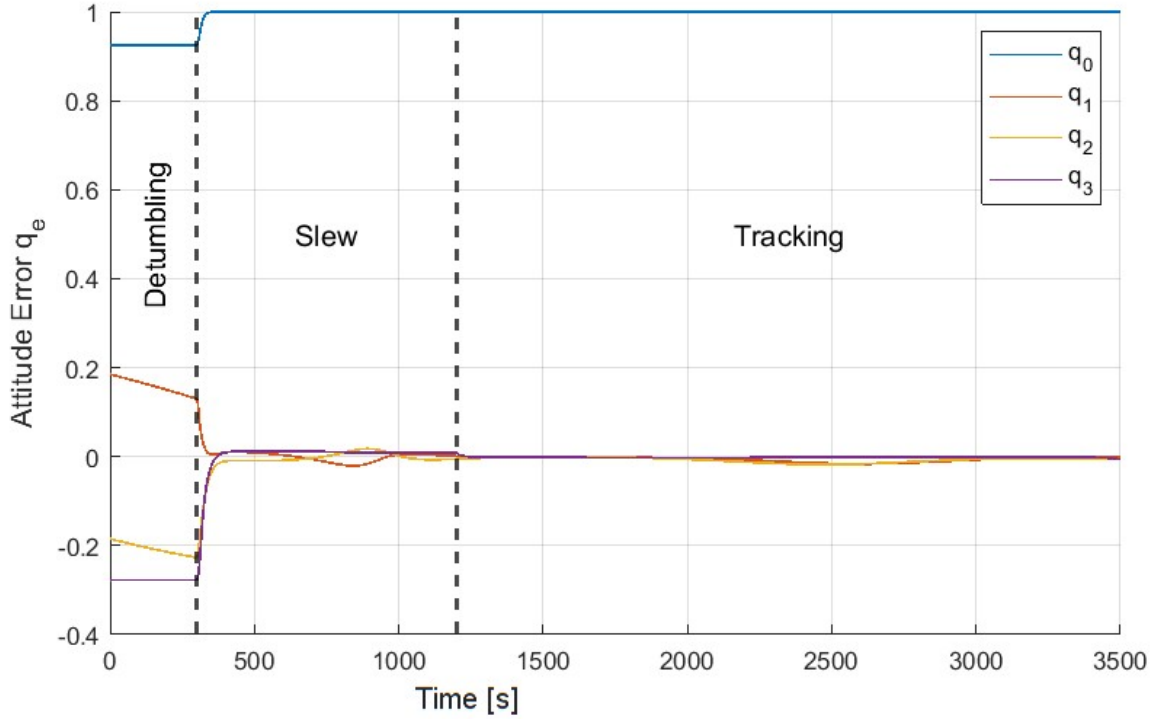


Figure 9: Attitude error throughout the different phases of orbiting.

Looking at the graph we see that the quaternion error approaches [1 0 0 0], which means that the rotation needed to go from the actual attitude to the desired one is small, as the scalar part is the cosine of half of the angle. This means that the attitude error is small enough.

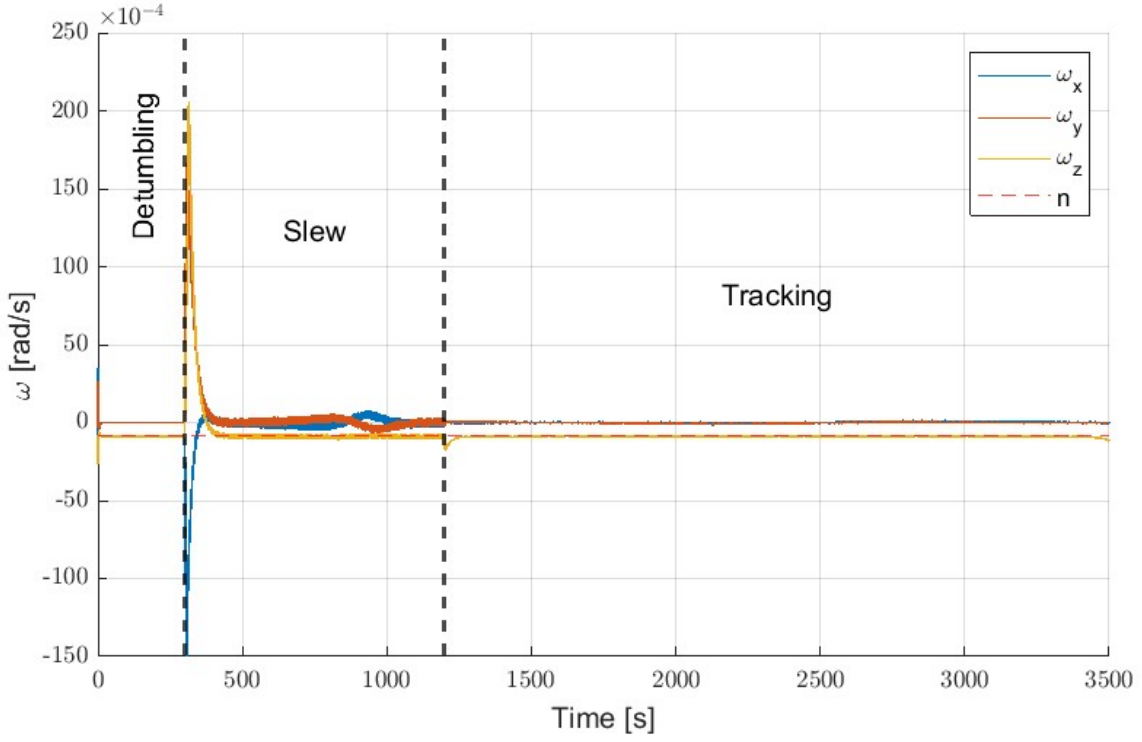


Figure 10: Angular velocity throughout the different phases of orbiting.

The control system managed to keep the angular velocities at their required rates, as shown, the z-axis angular velocity always settles at $-n$, which is the imposed required value, while the other two angular velocities are set as zero.

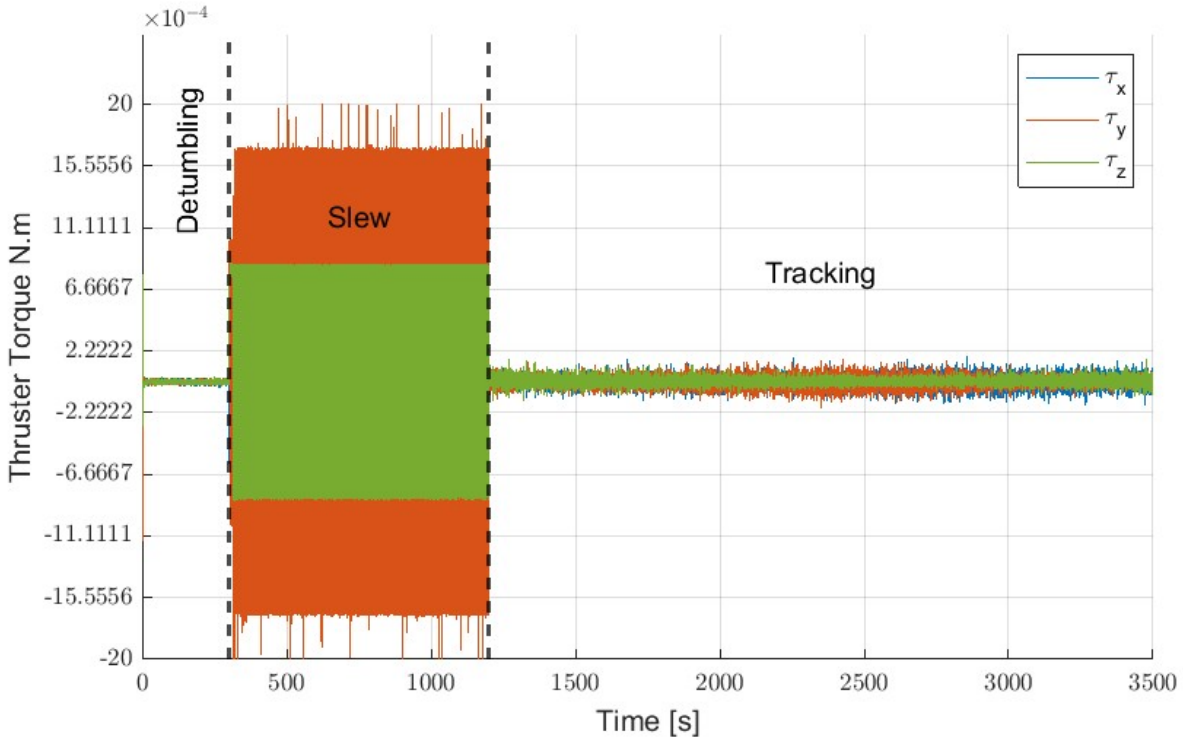


Figure 11: Torque generated from the thrusters throughout the different phases of orbiting.

This graph shown that the torques applied are always kept in the working zone of the thrusters. We can add a torque reach the maximum tolerated value for the slew maneuver on the y-axis, but it is correctly handled by the software and doesn't prevent the spacecraft to reach the commanded attitude.

5 Conclusion

The Simulink program handles three different mode for controlling the spacecraft attitude : de-tumbling, slew maneuver and Inertia Pointing.

The code works smoothly and provides results which are consistent with the theory : quaternions are under the good form and well normalized, angular velocity evolves as expected, actuators also works smoothly and remains in their working range condition.

The environment in which the spacecraft evolves is also well computed and well took into account.

We are proud to say that the project meets all the defined goals. This work allowed us to gain a huge experience in Matlab Simulink and was a good way to put into practice the attitude control theory.

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